



Process simulation and optimization of methanol production and utilization for electricity generation

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ABSTRACT

The utilisation of liquid organic carriers for hydrogen storage has demonstrated to be a highly effective method for storing hydrogen under ambient temperature and pressure conditions. Methanol has a hydrogen (H₂) content of 12.6% by weight and can be synthesised by the process of carbon dioxide (CO₂) hydrogenation. The main goal of the study is to demonstrate methanol's role as a hydrogen carrier and its application in a direct methanol fuel cell for energy generation. The suggested system synthesises methanol by utilizing green hydrogen with CO₂ recovered from industrial sources. The Aspen Plus simulation of the DMFC showed a 58% methanol conversion and a 61% energy efficiency (Methanol-to-power).

Introduction

With the ever-increasing demand for energy, there has been an upsurge in the use of oil, coal, natural gas, and various other natural resources. It causes depletion in their availability and causes a serious threat to the environment by releasing greenhouse gas emissions into the atmosphere, thus raising the earth's average surface temperature. This necessitates the need to exploit ways to utilize sustainable and innovative methods to replace fossil fuels as an energy source. Further, it also makes an important requirement to find energy-efficient systems that do not harm the environment and reduce operating costs. Sustainable, environmentally friendly sources include biomass energy, hydro power, nuclear energy, solar energy, and wind energy, which can very well replace fossil fuels. Given that it outperforms other renewable energy sources in terms of accessibility, affordability, capacity, and efficiency, the solar industry would unquestionably be the best choice for meeting future energy needs (W. H. Wang et al., 2015). Along with the naturally available sustainable options, commercially produced hydrogen plays a crucial role in bringing about the energy transition in the coming years. However, hydrogen storage presents a challenge in terms of volumetric and gravimetric energy densities that can be achieved at practical pressures compared to other gases (Broom & Hirscher, 2016). It also requires cryogenic storage since it has a boiling point close to 20.4 K White et al. (1950). One of the promising ways to store

hydrogen is the use of liquid organic hydrogen carriers like formic acid, methanol, and methyl cyclohexane, as they provide ease of handling and transportation (Onishi et al., 2018). Storing hydrogen in bound form as molecules liquid organic carriers have an energy efficiency of about 69 % and 88 % with and without energy recycling respectively (Wang et al., 2016). Recent advancements have enabled the use of CO₂ as a carbon source, integrating methanol production with existing infrastructure. "Green" methanol can be utilized across various sectors, including as a direct fuel, a chemical feedstock, and a means to generate electricity, thus offering versatile applications that align with the existing energy and chemical networks (Räuchle et al., 2016). Methanol is a promising liquid hydrogen storage medium due to its high storage and transit efficiency of 49 % at ambient temperature. (Räuchle et al., 2016). Methanol has an advantage over formic acid since it has a higher hydrogen storage capacity, thus carrying more hydrogen per unit mass or volume, higher energy density, lower manufacturing cost and longer shelf life. The energy density if methanol is 6400 Wh/kg (Shamsul et al., 2014) by mass as compared to energy density of formic acid (1725 Wh/kg) (Mardini & Bicer, 2021b). Methanol contains approximately 12.6 % hydrogen by weight, while formic acid contains about 4.4 % hydrogen by weight (Mardini & Bicer, 2021a; Shen et al., 2017). Methanol contains 65.22 % more hydrogen than formic acid. The ability to produce methanol at competitive prices and the potential for significant integration into industrial processes underscore its viability as a sustainable

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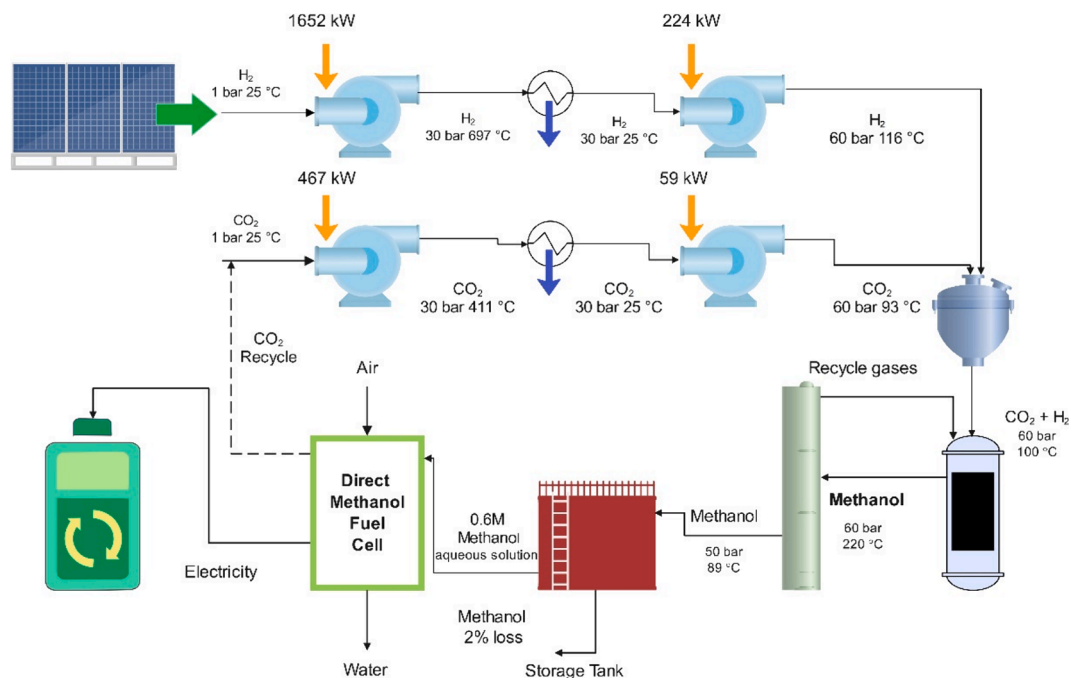


Fig. 1. The diagram illustrating the complete methanol production, storage, and utilization.

energy storage solution. The overview of methanol production, storage and utilization is shown in the Fig. 1.

Carbon dioxide (CO₂) hydrogenation reactions are crucial in producing chemicals liquid hydrogen carriers like methanol, ethanol, formic acid (Leonzio et al., 2019). India's carbon dioxide emissions increased from 615 metric tons (Mt) in 1990 to 1897 Mt in 2013, representing a 5 % compound annual growth rate. As a result, India's CO₂ emissions accounted for 2.7 % of world CO₂ emissions in 1990, rising to nearly 5 % in 2013 (Garg et al., 2017). CO₂ captured from these industries can be used as a starting material in the hydrogenation reaction for producing methanol, which in turn can be utilized for generation of electricity in a fuel cell.

Fuel cells, which emerged in the later part of the 20th century, are a contemporary form of energy converters that efficiently transform chemical energy into electrical energy. A fuel cell can persist in generating energy as long as fuel and oxidizer are available, as opposed to a typical battery, which eventually runs out of power (Lu et al., 2020). The kind of electrolyte, operating temperature, fuel used, and applications are a few of the variables that determine how fuel cells are classed. Direct Methanol Fuel Cell (DMFC) is similar to the polymer electrolyte membrane (PEM) both having an electrolyte in the form of a polymer membrane. However, the DMFC does not require a fuel reformer because the anode catalyst itself pulls the hydrogen from the liquid methanol. In a DMFC, the chemical reaction between methanol and oxygen results in the production of carbon dioxide (CO₂) and water (H₂O), which in turn generates electricity. DMFC are of two types, active and passive. The active DMFC system, however, tends to be more complicated and larger, having various additional components for fuel recirculation and phase separation (Mallick et al., 2015). An active system of DMFC is suitable for portable applications due to its high-power output (Mardini and Bicer, 2021a). The passive DMFC possesses a compact and uncomplicated design, as it relies on passive mechanisms such as diffusion to remove both the reactants and products. (Mallick et al., 2015; Zhao et al., 2009). Electronic gadgets such as Laptops, phones, and other portable applications are the areas where DMFC is used (Pratschner et al., 2021). Methanol can be stored in conventional energy storage devices like fuel cells and batteries, providing a flexible and scalable means to store and deploy renewable energy on-demand. Additionally, methanol is well-suited for use in DMFCs, which are

compact and efficient for portable and stationary power applications (Räuchle et al., 2016).

Multiple studies have possibility to capture carbon dioxide (CO₂) and use it as a raw material for hydrogenation reactions. This process can be used to make liquid hydrogen carriers, which can be used for storing hydrogen. Out of all these carriers, methanol is particularly notable for its enormous storage capacity and efficient energy transfer qualities. Nevertheless, the utilization of methanol as a fuel for energy production, namely in terms of heat and electricity, is still restricted.

In this paper, methanol is produced from CO₂ hydrogenation by compressing CO₂ and H₂. The study aims to study the energy utilization by a methanol production plant integrated with a direct methanol fuel cell. Additionally, heat integration process is developed to optimize energy cost. The objective of heat integration techniques is to minimize the use of heating and cooling resources by aligning process streams for efficient heat recovery. Methanol production is modelled in ASPEN PLUS®. Methanol is further used in a DMFC which converts methanol and oxygen into carbon dioxide, water, and power in the form of electricity. CO₂ is recycled back to the system to produce methanol, and H₂O can be used for hydrogen generation using electrolysis. The objective of this study is to investigate and evaluate the feasibility of utilizing methanol in fuel cells for the purpose of generating electricity, emphasizing its potential as a substitute energy resource.

Materials and methodology

Methanol production unit

The system comprises four integrated subsystems: methanol production from CO₂ and H₂, fuel cell, photovoltaic (PV)-assisted electrolysis, and PV power plant. In this work, it is assumed that the power is generated through a PV system which is used in production of methanol unit, specifically in compressing the CO₂ and H₂ gases. Equation (1) can be used to determine the area of the solar panel.

$$W_{in} = \eta_{pv} I A_{pv} \quad (1)$$

Here, I , (526 W/m²) (Kamarudin et al., 2009) is the average solar irradiation (power per unit area received from the sun in terms of

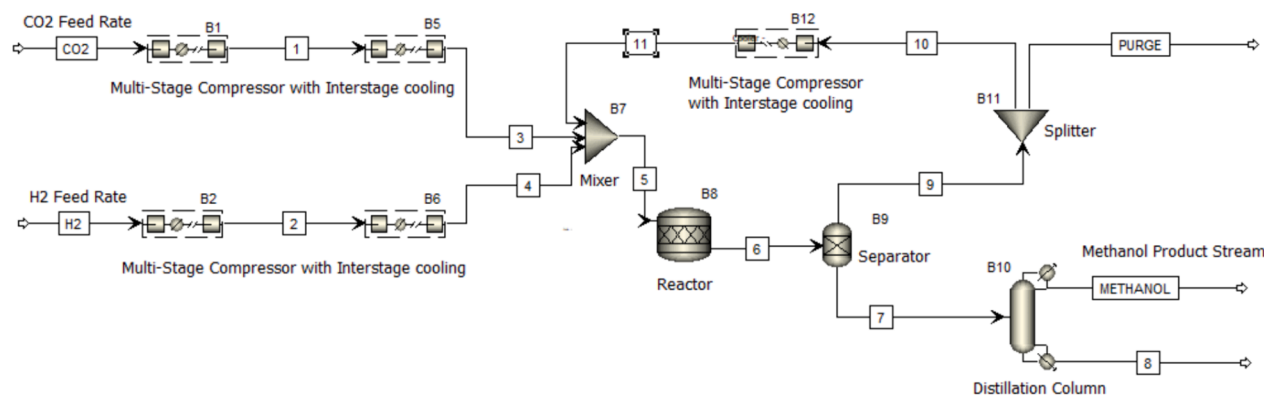


Fig. 2. A process flow diagram for methanol manufacturing created using Aspen Plus.

electromagnetic radiation), while η_{pv} and A_{pv} , respectively, are the PV panels' efficiencies and areas (Mardini & Bicer, 2021a). CO₂ in the feed stream (100 kmol/hr) at ambient temperature and the pressure undergoes compression in two phases, first from 1 bar to 30 bar at a pressure ratio of 3.10, and subsequently from 30 bar to 60 bar at a pressure ratio of 1.26. Each phase employs a 3-stage compression. Employing a two-phase compression method enhances volumetric efficiency by decreasing the pressure ratio at each phase and reducing the temperature at the end of each compression cycle. H₂ obtained from PV-assisted electrolysis of water, is compressed to 60 bar in a two-phase process from ambient temperature and pressure. H₂ enters the compressor at a flow rate of 300 kmol/hr. The power required for the compression of CO₂ and H₂ is assumed to be obtained from the PV power plant. During the compression phase, coolers are utilised to decrease the temperature of the feed stream. The Aspen Plus process model for the production of methanol is shown in Fig. 2.

Several studies have demonstrated the utilization of membrane reactors for the synthesis of methanol from carbon dioxide (CO₂) and hydrogen (H₂). A study demonstrated the utilization of a membrane reactor that exhibits permeability to water and methanol (Atsonios et al., 2016). The findings revealed that the incorporation of membranes enhances the production of methanol. A different study manipulated the H₂/CO₂ ratio and saw greater methanol selectivity and yield in a zeolite membrane reactor compared to a conventional reactor (Gallucci et al., 2004). Gallucci and Basile conducted a study where they simulated a membrane reactor using a commercially available catalyst (Gallucci & Basile, 2007). They discovered that the membrane reactor achieved greater levels of CO₂ conversion and methanol selectivity compared to a traditional reactor, under identical experimental conditions. Moreover, methanol production often involves the utilization of adiabatic reactors arranged in a multi-bed configuration with inter-stage heat exchange, as well as multi-tubular reactors of the Lurgi type (Bozzano & Manenti, 2016). These reactors efficiently dissipate the heat produced during the reaction by utilizing externally circulated boiling water for cooling purposes. A stoichiometric reactor is employed in our work to describe methanol production. This type of reactor enables direct control over reactant conversion, hence simplifying the process simulation. This method allows for precise calculations of material and energy quantities, which helps in designing and optimizing processes effectively, especially in initial feasibility studies.

For methanol synthesis, the pressurized streams of CO₂ and H₂ are mixed, and then the mixed stream with a flow rate of 400 kmol/hr and a temperature of 98.85 °C enters the reactor to produce methanol and water according to equation 2. The reaction takes place at 60 bar and 220 °C with a fractional conversion of 17.25 % CO₂. The catalyst used in this process is Cu/ZnO/Al₂O₃ catalyst.



CO₂ hydrogenation being an exothermic reaction. As a result, heat causes the temperature to rise to 220 °C. The product stream passes through a flash drum where the pressure is reduced to 50 bar and the corresponding temperature is reduced to 5 °C. CO₂ and H₂ are recycled back to the conversion reactor with a recycle ratio of 80 %, and the methanol water stream sent to the distillation column. In the distillation column, liquid phase and vapor phase interact with each other on each tray with the assumption that they attain equilibrium giving maximum tray efficiency. An equilibrium-based approach is used for simulating the distillation column.

The model is built upon the subsequent assumptions. The system operates at a constant temperature and remains in a stable condition. The system is simplified to a zero-dimensional model and any pressure drop is disregarded. Production of methanol was modelled on Aspen Plus employs the PENG-ROB fluid package, specifically designed for nonpolar or slightly polar mixes, such as hydrocarbons or light gases like

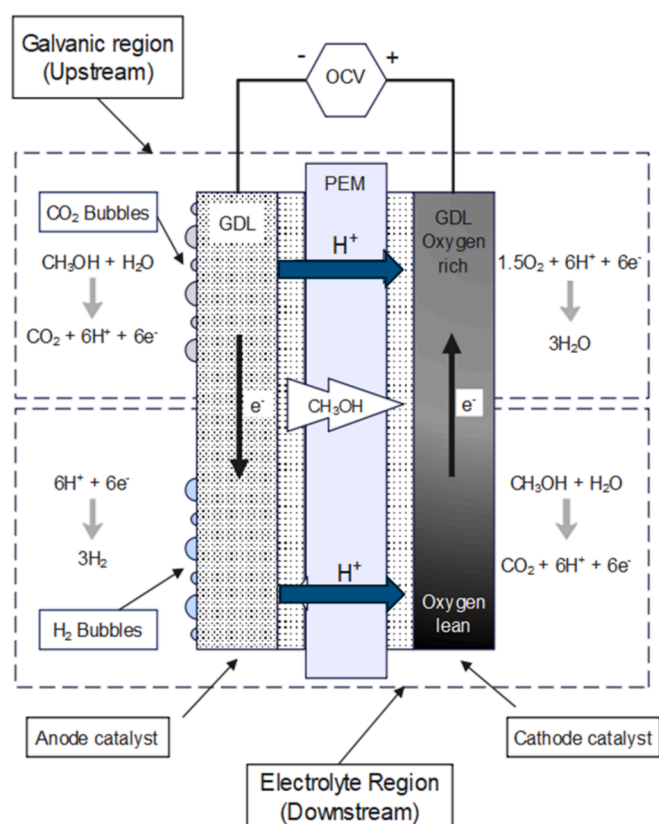


Fig. 3. Direct methanol fuel cell.

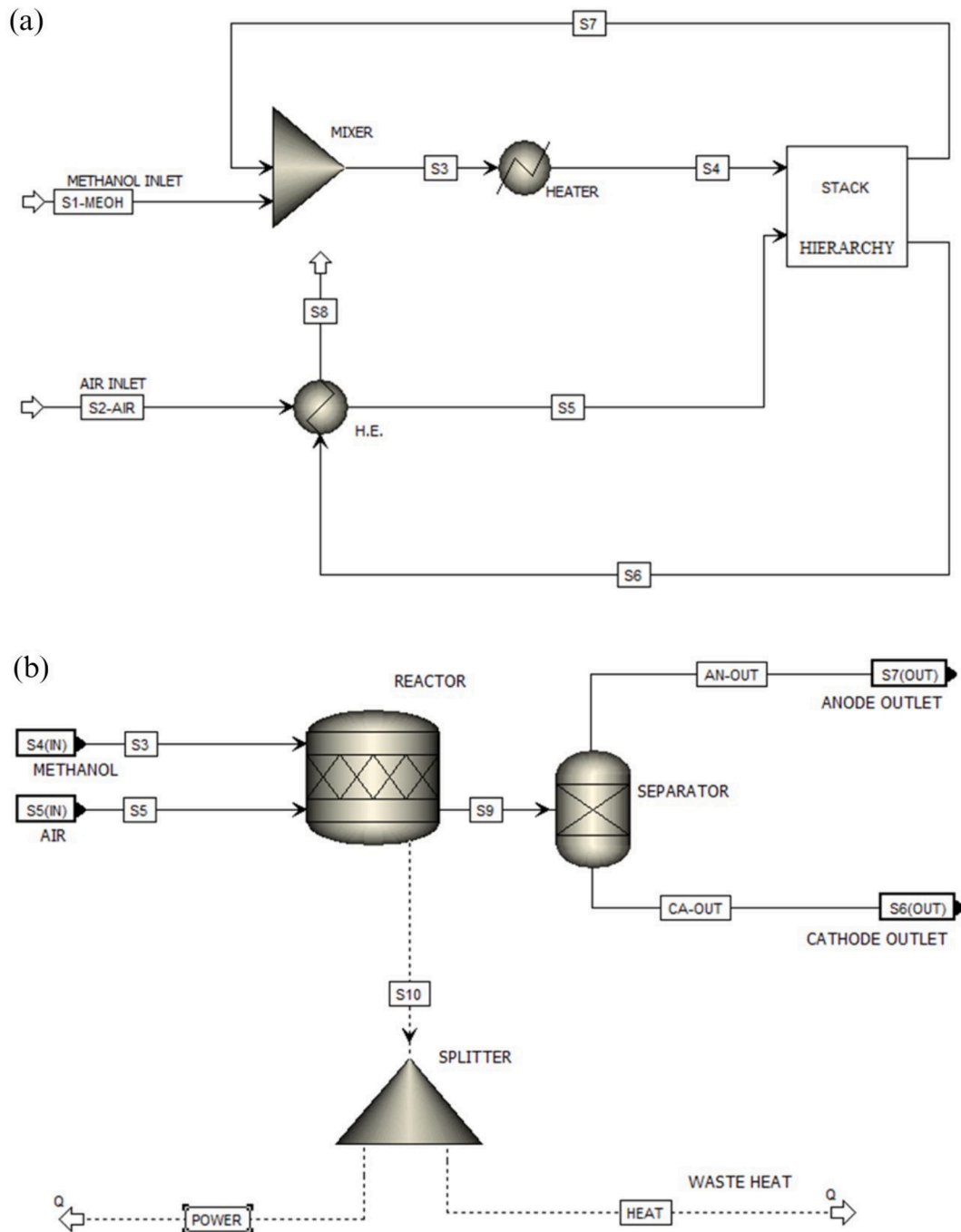


Fig. 4. (a) Fuel cell model. (b) Methanol fuel cell stack.

carbon dioxide and hydrogen.

- The compressor's isentropic efficiency is 72 % (Räuchle et al., 2016).
- Conversion reactor used for production of methanol works with 17.25 % of CO₂ conversion and the pressure is 60 bar (Lee et al., 2017).
- The thermal power output of the conversion reactor is 1162 kW which is obtained from the simulation.

The following is a list of the primary phases of the integrated system:

- Hydrogen generated from PV assisted electrolysis along with CO₂ used for synthesizing methanol.

- The necessary quantity of electricity for compression and electrolysis is produced utilizing photovoltaic systems.
- Conservation of energy in methanol production using heat integration techniques.
- The total amount of hot and cold utility required for the process is calculated.

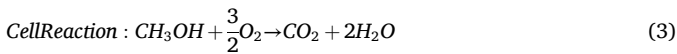
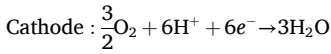
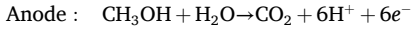
Direct methanol fuel cell

Methanol undergoes decomposition within the fuel cells, resulting in the generation of electricity and H₂O. The resulting CO₂ and H₂O are subsequently recycled to produce methanol and H₂. This makes this system a net-zero CO₂ emissions process.

Methanol obtained from CO₂ hydrogenation is used as fuel to

generate electricity in a DMFC as shown in Fig. 3. Without any external reformulation of methanol into hydrogen gas, the fuel cell is specifically made to directly utilize methanol. At anode, oxidation of methanol takes place to produce protons (H^+ ions), electrons (e^-), and carbon dioxide (CO_2). An electrolyte like potassium hydroxide (KOH) or sodium hydroxide (NaOH) allows the migration of protons from anode to cathode while preventing electron flow. At cathode, oxygen (from air) reacts with the released electrons and protons to form water (H_2O).

The electrons flow through an external circuit from anode to cathode and create an electric current. The overall cell reaction is given by equation (3):



DMFC is simulated in Aspen Plus as shown in Fig. 4. Here, the overall oxidation of methanol is considered since Aspen Plus is unable to simulate an ion crossover through fuel cells. Methanol enters the fuel cell at ambient condition and a flow rate of 1553 kg/hr. CO_2 and H_2O generated during the reaction are sent back to the methanol production plant. The model is based on the following assumptions: isothermal and steady state operation, zero dimensional, all working fluid treated as ideal, and pressure drop neglected. Methanol crossover is a notable concern in DMFC, as it involves the passage of methanol molecules through the membrane from the anode to the cathode without undergoing the required electrochemical reaction (Alias et al., 2020). The permeation of methanol across the membrane results in the loss of fuel, decreased efficiency, and the possibility of catalyst contamination at the cathode. The literature extensively documents the occurrence of methanol crossover, which has a severe impact on the overall performance and efficiency of DMFCs. In contrast to other types of fuel cells, DMFCs generally do not experience water crossover (Zhao & Xu, 2009). Instead, the main issue that needs to be tackled to improve the practicality and effectiveness of DMFC technology is methanol crossover.

Methanol obtained from CO_2 hydrogenation which is stored at ambient temperature and pressure is (mixed with recycled methanol stream from the anode) heated to 65 °C. Air is used as the source of oxygen and heated to 63 °C. The two streams are introduced in the reactor, where they undergo an electrochemical reaction resulting in simultaneous generation of heat and power. The total energy output from this process amounts to 9098 kW. This energy is generated in the form of electric current. A direct methanol fuel cell has a 97 % theoretical energy conversion efficiency (Scott and Xing, 2012), (Shamsul et al., 2014) and the rest is waste heat which can be recovered. Unreacted methanol is recycled into the feed stream and the stream containing CO_2 , H_2O , and unreacted O_2 is used to heat air coming in the feed. These products are used in methanol production, thus making the process a net-zero CO_2 emission process.

$$\eta_{\text{fuel}} = \frac{\dot{m}_{\text{an,cons,M}}}{\dot{m}_{\text{an,cons,M}} + \dot{m}_{\text{an,cons,M}}^{\text{cross}}} \times 100 \quad (4)$$

Here, $\dot{m}_{\text{an,cons,M}}$ = Methanol consumption associated with anodic reaction (g/s)

$\dot{m}_{\text{an,cons,M}}^{\text{cross}}$ = Methanol consumption on cathode side due to crossover (g/s)“.

Three primary assumptions made for carrying out the simulation are as follows:

1. Fuel efficiency is assumed to be 58.3 %. Hence, methanol available at anode for consumption is 251.4 g/s (Shen et al., 2017).

Table 1

Stream properties of methanol production unit.

Stream Name	Temperature	Pressure	Molar Vapor Fraction	Molar Liquid Fraction
Units	C	bar		
1	25	30	1	0
2	25	30	1	0
3	93	60	1	0
4	116	60	1	0
5	100	60	1	0
6	220	60	1	0
7	219	40	1	0
8	150	5	0	1
9	219	40	1	0
10	219	40	1	0
11	100	60	1	0
CO_2	25	1	1	0
H_2	25	1	1	0
METHANOL	110	5	0	1
PURGE	219	40	1	0

Table 2

Mass balances of all stream.

Stream Name	Mass Flows	CO_2	H_2	METHANOL	WATER
Units	kg/hr	kg/hr	kg/hr	kg/hr	kg/hr
1	4401	4401	0	0	0
2	605	0	605	0	0
3	4401	4401	0	0	0
4	605	0	605	0	0
5	17,146	14,837	2039	173	97
6	17,146	12,277	1687	2036	1145
7	2863	0	0	1833	1030
8	1039	0	0	18	1020
9	14,283	12,277	1687	204	114
10	12,140	10,436	1434	173	97
11	12,140	10,436	1434	173	97
CO_2	4401	4401	0	0	0
H_2	605	0	605	0	0
METHANOL	1825	0	0	1815	10
PURGE	2142	1842	253	31	17

2. This study assumes that 91 % of the methanol that crossed over to the cathode underwent an oxidation reaction, while the other 9 % evaporated without reacting and was discharged from the cathode. These assumptions are based on the findings of elemental experiments conducted with a single cell (Shen et al., 2017).

Results & discussion

Methanol production unit

Temperature, pressure and mole fractions of liquid and vapor stream is given in Table 1 Stream Properties of methanol production unit. Table 2 provides the predicted mass flow rates for all the streams, taking into account the assumed feed of CO_2 and H_2 . CO_2 feed rate is assumed to be 100 kmol/hr and fed at ambient temperature and pressure condition and H_2 flow rate is taken as 300 kmol/hr according to the stoichiometry at ambient temperature and pressure condition. Based on the feed condition and other assumptions, the plant has a capacity of producing 1555 kg/hr (48.5 mol/hr) of methanol. The densities and enthalpies of the streams employed in the production of methanol are displayed in Table 3 and Table 4.

The PV energy generation with an efficiency of 20 % is assumed to deliver 2.402 MW of power to compress the feed gas to a pressure of 60 bar utilizing a two-stage compression cycle. The area necessary for the PV panels is determined to be 22,832 m^2 based on the energy needs of the compression system. Inter-stage coolers are used to cool the compressed CO_2 and H_2 stream before they undergo a second compression. The distillation column is designed to have 30 stages with reboiler at

Table 3

Enthalpy, entropy and densities of all streams.

Stream Name	Molar Enthalpy	Molar Entropy	Molar Density	Mass Enthalpy	Mass Entropy	Mass Density	Enthalpy Flow
Units	kJ/kmol	kJ/kmol-K	mol/l	kJ/kg	kJ/kg-K	gm/ml	kW
1	-3.95E+05	-2.86E+01	1.48E+00	-8.97E+03	-6.51E-01	6.51E-02	-1.10E+04
2	1.74E+01	-2.82E+01	1.19E+00	8.64E+00	-1.40E+01	2.40E-03	1.45E+00
3	-3.93E+05	-2.70E+01	2.38E+00	-8.93E+03	-6.12E-01	1.05E-01	-1.09E+04
4	2.70E+03	-2.63E+01	1.81E+00	1.34E+03	-1.30E+01	3.65E-03	2.25E+02
5	-9.71E+04	-2.22E+01	1.91E+00	-7.70E+03	-1.76E+00	2.41E-02	-3.67E+04
6	-1.05E+05	-1.83E+01	1.45E+00	-7.58E+03	-1.33E+00	2.00E-02	-3.61E+04
7	-2.27E+05	-1.18E+02	2.08E+00	-9.09E+03	-4.71E+00	5.22E-02	-7.23E+03
8	-2.77E+05	-1.39E+02	4.72E+01	-1.52E+04	-7.66E+00	8.57E-01	-4.40E+03
9	-9.35E+04	-9.65E+00	9.67E-01	-7.39E+03	-7.62E-01	1.22E-02	-2.93E+04
10	-9.35E+04	-9.65E+00	9.67E-01	-7.39E+03	-7.62E-01	1.22E-02	-2.49E+04
11	-9.75E+04	-2.24E+01	1.91E+00	-7.71E+03	-1.77E+00	2.42E-02	-2.60E+04
CO ₂	-3.94E+05	2.90E+00	4.06E-02	-8.94E+03	6.59E-02	1.79E-03	-1.09E+04
H ₂	5.51E-01	1.07E-01	4.03E-02	2.73E-01	5.32E-02	8.13E-05	4.59E-02
METHANOL	-2.32E+05	-2.19E+02	2.14E+01	-7.26E+03	-6.86E+00	6.82E-01	-3.68E+03
PURGE	-9.35E+04	-9.65E+00	9.67E-01	-7.39E+03	-7.62E-01	1.22E-02	-4.40E+03

Table 4

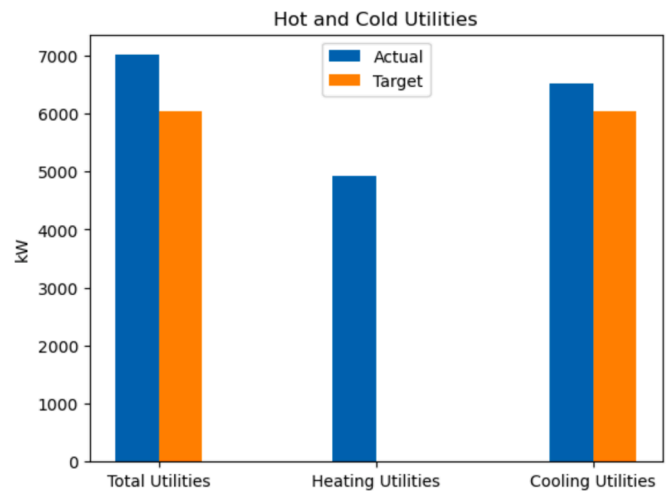
Mass flow rate, enthalpy, work for the reactor, distillation column, compressor and cooler.

Equipment	Specific Enthalpy in (kJ/kg)	Specific enthalpy out (kJ/kg)	Q (kW)
Conversion Reactor	-7.67E+03	-7.55E+03	4.92E+02
Compressor 1	-8.94E+03	-8.56E+03	3.58E+02
Compressor 2	2.73E-01	9.83E+03	1.16E+03
Compressor 3	-8.97E+03	-8.93E+03	5.90E+01
Compressor 4	8.64E+00	1.34E+03	2.3E+02
Compressor 5	-7.84E+03	-7.21E+03	5.6E+02
Inter-stage coolers in Compressor 1	—	—	-3.97E+02
Inter-stage coolers in Compressor 2	—	—	-1.15E+03
Inter-stage cooler in Compressor 5	—	—	-1.62E+03
Flash Drum	-7.55E+03	-1.84E+04	-3.12E+03
Distillation Column	-1.06E+04	-2.26E+04	Condenser: -1.16E+03 Reboiler: 1.4E+03

Table 5

Process modelling results compared to literature.

Parameter	Literature	Present model results
Methanol Formation rate	Methanol: 0.388 mol/hr (Lee et al., 2017) Feed required: 2.28 mol/hr of CO ₂	Methanol: 1555 kg/hr (48.5 mol/hr) Feed required: 100 kmol/hr of CO ₂ 300 kmol/hr of H ₂
Specific Power consumption	10.76 kWh/kg methanol produced (Pratschner et al., 2021)	8.24 kWh/kg methanol produced
Methanol fuel cell	MeOH use: 0.95 g/min Fuel utilization: 58.3 % Stack Power output: 131 W Temp: 65 °C (Shen et al., 2017)	MeOH use: 1525 kg/hr (2 % storage loss = 30 kg) 58 % fuel utilization Power Output: 2038.3 kW Temperature: 65 °C
Methanol reactor efficiency	Recycle Ratio: 80 % Reaction Temperature: 220 °C MeOH Yield in reactor: 17.25 % MeOH Yield in whole process: 61.1 % (Leonzio et al., 2019)	Recycle Ratio: 80 % Reaction Temperature: 220 °C MeOH Yield in reactor: 14 % MeOH Yield in whole process: 48.5 %
DMFC efficiency	21.6 % (Shen et al., 2017)	61 % (Based on the simulation)

**Fig. 5.** Utilities present in CO₂ hydrogenation model.

151 °C and condenser at 89 °C and 5 bar operating pressure. The distillation column has a distillate rate of 49 kmol/hr reflux ratio of 1.35. The heat duty of the reboiler and condenser in the distillation column is 1396 kW and 1158 kW respectively. In this process, the recycle ratio is kept at 80 %, thus getting an overall methanol yield for the process as 48.5 %. Table 5 demonstrates that the methanol yield falls by 12 when compared to literature. However, the direct methanol fuel cell achieves an efficiency of 61 %.

By enhancing heat recovery and reducing the need for external utilities, heat integration is used to lower the plant's energy costs. The Aspen Energy Analyzer20 is used to design the best heat exchanger network and retrofit. With the help of this tool, energy waste can be reduced, resulting in a more affordable and environmentally friendly procedure. Aspen Energy Analyzer analyses the facility and makes recommendations for improvements that can save energy expenses.

The heating utility has a total amount of 491 kW, whereas the cooling utility has a total amount of 6521 kW, as seen in Fig. 5. The concept of saving potential quantifies the disparity between the present value and the target value, providing insight into the extent of potential cost reduction. The energy analyzer identified a single solution for the design of a new heat exchanger that connects cooler B4 to the conversion reactor. The heat exchanger comprises a fluid on the hot side derived from the cooler and a fluid on the cold side that comes from the reactor. The implementation of a novel heat exchanger design has the potential to yield annual savings of around \$80,283, while requiring a capital investment of \$14,808. The comparison between the original and

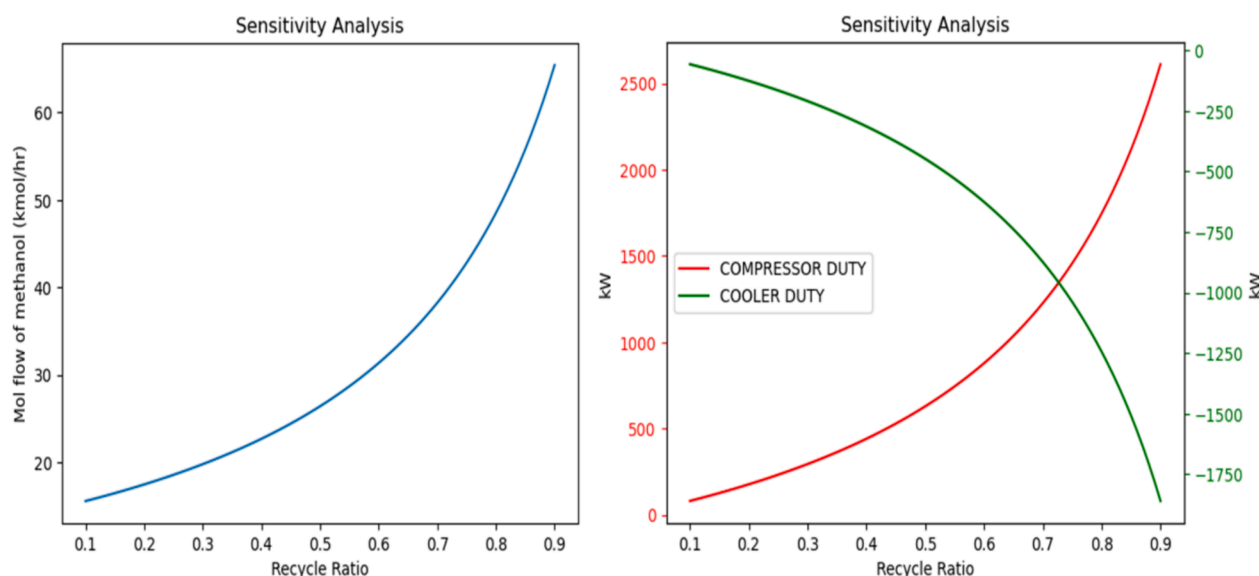


Fig. 6. Sensitivity analysis.

retrofit scenarios demonstrates that the latter results in a decrease in energy expenses, accompanied by an increase in initial investment costs. However, this increase in capital costs can be compensated for within a period of 0.2 years. It should be noted that the ratio \$14,808/\$80,283/year is equal to 0.2 years, referring to the duration needed to recover the extra investment expenses. A total energy saving of 13.85 % can be achieved after a period of 1 year.

The methanol yield of the current system, excluding the incorporation of a recycle stream, is 14 %. An investigation was carried out to examine the influence of recycling on the production of methanol. Sensitivity analysis is used to examine the influence of different values of an independent variable on a certain dependent variable, while maintaining a fixed set of assumptions. A sensitivity analysis was conducted on the methanol processing system by varying the recycle ratio from 0.1 to 0.9. The impact of this adjustment on the mol flow of methanol in the product stream was noted. The methanol flow demonstrated exponential increase, starting at 15 kmol/hr with a recycle ratio of 0.1 and reaching 65.4 kmol/hr with a recycle ratio of 0.9. An additional compression and cooling system is necessary to handle a recycle stream, which leads to an increased energy demand. The energy demands of the compressor and cooler are illustrated in Fig. 6. The energy requirement of the compressor experiences exponential rise with the increase in recycle rates. Hence, it is crucial to select an ideal recycling ratio that may offer a satisfactory output and efficient energy utilisation. The chart indicates that the ideal recycling ratio for this study ranges from 0.7 to 0.8.

When comparing literature with the current model results as shown in Table 5, significant disparities emerge in methanol formation, methanol fuel cell performance, reactor efficiency, and DMFC efficiency. In the current model, the rate of methanol synthesis is considerably larger at 1,555 kg/hr (48.5 mol/hr) in comparison to the 0.388 mol/hr described in the literature. This increase is linked to varying feed demands: the model necessitates 100 kmol/hr of CO₂ and 300 kmol/hr of H₂, while the literature states 2.28 mol/hr of CO₂. The specific power consumption for producing 1555 kg/hr of methanol is 8.24 kWh/kg of methanol produced which is 23.42 % less than found in another study (Pratschner et al., 2021). This can be attributed to the multi-phase operation of the compression cycle. It is to be noted that the energy demand for H₂ which comes from electrolysis is not considered. The model for the methanol fuel cell indicates a methanol consumption rate of 1,525 kg/hr, taking into account a storage loss of 2 %. The fuel utilization rate remains consistent at 58 %. Nevertheless, the power output in the model is far more at 2,038.3 kW, in contrast to the literature's 131

W, despite both operating at a temperature of 65 °C. The reactor in the model maintains the same recycle ratio and reaction temperature as stated in the literature (80 % and 220 °C, respectively). However, the methanol production is somewhat lower in the reactor (14 % compared to 17.25 %) and in the overall process (48.5 % compared to 61.1 %). The DMFC efficiency (methanol-to-power) in the model is significantly greater at 61 % compared to the 21.6 % reported in the literature.

Fuel cell

An Aspen Plus simulation is conducted to describe the complete oxidation of methanol, as the phenomenon of ion crossover via the fuel cell cannot be accurately represented in Aspen Plus. Methanol obtained from CO₂ hydrogenation and air are heated using two heat exchangers. Methanol stream is mixed with the recycle stream of methanol and is heated in a heater to 65 °C and the products obtained from the electrochemical reaction in the fuel cell are used as a heating fluid in heat exchanger to heat air. In a fuel cell methanol oxidation takes place at anode and the electrons recombine with oxygen and hydrogen to form water at cathode. The simulation uses a conversion reactor operating at 65 °C and 1 bar. 58 % conversion of methanol is obtained with an energy generation of 9098 kW. In addition, the simulation showed that the DMFC had an energy efficiency of 61 %, stating that 61 % of energy derived from methanol is converted into usable energy. An energy output of 9,098 kW was produced. Given that a typical DMFC exhibits an energy conversion efficiency ranging from 30 % to 40 %, it is estimated that approximately 2,730 kW of electrical energy can be generated from this system. The remaining energy, which is not converted to electricity, is dissipated as waste heat. This thermal energy can potentially be recovered and utilized in other applications, enhancing the overall efficiency of the process.

A pure CO₂ and H₂ stream are taken as feed to produce methanol with a yield of 48.9 % at a rate of 1555 kg/hr. The energy required during the compression of the gas streams in this process is to be obtained from a photo-voltaic system with an efficiency of 20 % and spread over an area of 22,858 m². The other unit operations which include the reactor, and the distillation column require approximately 600 kW of energy. The methanol converted in DMFC to generate electricity and waste heat which can be recycled back within the process. Overall, half of the energy requirement of the process is satisfied from the renewable source of PV plant and in turn the process also generates energy in the form of heat and electricity.

Conclusion

The increase in the demand for energy has forced to investigate sustainable and renewable ways to produce and store energy. Solar energy plays a fundamental role to produce electricity and along with this, energy from hydrogen will also play a crucial role. In the present work, energy from the photovoltaic (PV) system is utilized in synthesizing methanol which acts as a liquid organic hydrogen storage material with high energy density and contains high H₂ content. Methanol is further used in Direct Methanol Fuel Cell to undergo an electrochemical oxidation reaction to generate electricity, heat and H₂O and CO₂ which can be recycled back for generation of hydrogen and methanol respectively. The methanol production plant generates 1585 kg/hr of methanol which in turn can generate 2730 kW of electricity via fuel cell. By turning surplus electricity and industrial CO₂ emissions into methanol as an energy carrier, the suggested technology can enable large-scale renewable energy storage. Future work will involve the simulation of PV-based electrolysis, followed by the calculation of specific energy demands through the consideration of all relevant energy requirements. Additionally, a single cycle will be created to enable a comprehensive overall analysis.

CRedit authorship contribution statement

Vikram S. Chatake: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Harsh V. Rambhia:** Writing – original draft, Validation, Methodology, Formal analysis. **Sammed S. Bhalariao:** Validation, Methodology, Formal analysis, Data curation. **Vikram S. Korpale:** Validation, Methodology, Formal analysis, Data curation. **Jayendra P. Ahire:** Writing – review & editing, Visualization, Methodology, Data curation, Conceptualization. **Aniruddha B. Pandit:** .

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.wmb.2024.09.005>.

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